Causal Workbench: tutorial and viva defense

Your 60-second explanation

Our project separates language processing from formal causal reasoning. In Phase 1 we use classical machine learning. We first test whether a causal hypothesis follows from a Corr2Cause premise. Separately, we classify directed cause-effect relations in SemEval sentences with supplied entity spans. Candidate edges need human review before entering a DAG. The graph rejects cycles and self-loops. Given matching binary observations, we estimate Bayesian conditional probability tables. Finally, we compare observational queries with intervention queries in a controlled synthetic system. Our experiments show that text classification can exploit wording shortcuts, so we do not claim that a high F1 establishes causal understanding.

1. What enters and leaves each component?

Corr2Cause reasoning input is a premise and a hypothesis, output is entailed or not entailed. It does not return entity pairs. SemEval input is a sentence with two marked entity spans, output is no causal relation, e1 causes e2, or e2 causes e1. The lexical extraction demo takes plain text and returns candidate triples with sentence provenance. A reviewer approves edges. The graph stage produces an acyclic structure. The parameter stage additionally requires a table of observations. The query stage returns a probability conditional on the graph, fitted tables and assumptions.

Example: 'Rain causes wet grass.' yields rain -> wet_grass with a document ID and sentence ID. This states what a sentence says. To estimate the probability of wet grass, we need observations of rain and wet_grass. The text model's score cannot substitute for a CPD.

2. How does TF-IDF work?

A bag of words counts tokens without full syntax. TF-IDF reduces the importance of common words. Smooth IDF(t) = log((1+N)/(1+df(t))) + 1. Sublinear TF uses 1+log(count) for present tokens. L2 normalization scales the vector to length one. Bigrams preserve short phrases such as 'caused by'. Retaining single-letter tokens matters because Corr2Cause uses A, B and C as variables. Fit vocabulary and IDF on training text only. Test text must never influence feature fitting.

Logistic regression learns linear class scores. Binary reasoning uses sigmoid(w.x+b). Directed extraction uses a three-class logistic classifier. C is inverse regularization strength: small C gives stronger regularization. Balanced weights make rare classes matter more during fitting. The extraction experiment selects C from 0.1, 1 and 10 on development edge F1. The test set reports performance after that choice. This search does not guarantee a globally optimal model.

3. What is leakage, and what do our splits prevent?

Several Corr2Cause rows share one premise but ask different hypotheses. Random row splitting could put almost identical situations on both sides. We hash normalized premises and purge overlap across train, dev and test. This blocks exact premise reuse. It does not block isomorphic graphs under variable renaming or shared templates.

SemEval has multiple marked pairs in some sentences. We group by the underlying normalized sentence without span tags. All pairs in that sentence remain in one split. We deduplicate marked pairs separately, because two different pairs in one sentence can legitimately have different labels. Document IDs are unavailable, so we cannot honestly claim full document-disjoint evaluation.

4. Precision, recall and F1

Precision = TP/(TP+FP). Recall = TP/(TP+FN). F1 = 2PR/(P+R). For directed extraction, predicting B->A when gold is A->B creates one FP and one FN. A correct direction is essential. Macro F1 averages per-class F1 equally. Majority accuracy can look strong on imbalanced data while causal-relation recall is zero. Always name the task and metric before giving a number. Corr2Cause macro F1 and SemEval directed edge F1 are different measurements.

A bootstrap interval resamples evaluation units to show sampling variability conditional on the frozen fitted model. Corr2Cause resamples premise groups. The extraction run resamples sentence groups. Neither interval includes all variability from retraining, selecting hyperparameters or changing annotation policy.

5. Why the shortcut experiment matters

Full Corr2Cause input produces macro F1 about 0.700. A hypothesis-only model obtains about 0.653 without reading the premise. Therefore, premise-free wording cues support much of the observed score. Do not say this proves that every prediction is a shortcut, or interpret the difference as an exact causal contribution of the premise. The diagnostic challenges the strength of the reasoning claim.

The SemEval entity-only ablation asks a different question: can nominal strings predict the relation without surrounding context? The main extraction representation masks actual entity strings while preserving ENTITY1/ENTITY2 markers. This encourages use of surrounding context but does not guarantee genuine causal comprehension.

6. What does the graph add?

A directed acyclic graph has directed edges and no directed cycles. For the selected static Bayesian model, acyclicity enables a factorization into P(node | parents). A->B, B->C, C->A fails this contract. Rejecting the last edge ensures valid structure but does not show which edge is actually wrong. The current algorithm is greedy and source-order dependent. Human approval records responsibility and provenance, not ground-truth proof.

Real causal feedback can exist. It requires a time-indexed or other appropriate model rather than silently violating this static DAG representation. A graph can be acyclic and still causally false.

7. Where do probabilities come from?

A Bayesian network stores one conditional probability distribution per node given its parents. Our teaching system factors as P(R,S,W)=P(R)P(S|R)P(W|R,S). We generate 6,000 binary observations from a declared synthetic SCM. BDeu smoothing with equivalent sample size five adds prior mass to avoid brittle zero-frequency estimates. The reviewed-network path fits the same kind of tables to uploaded binary observations with matching node names.

Sparse parent combinations weaken estimates. Smoothing prevents zeros but does not create evidence. Causal interpretation needs correct structure, measured confounders, adequate support/positivity and stable mechanisms. An observational BN without defensible causal assumptions only supports probabilistic association.

8. Observation versus intervention, with numbers

Observation asks P(W=1 | S=1): among cases with sprinkler on, how often is grass wet? Intervention asks P(W=1 | do(S=1)): if we force the sprinkler on, how often would grass be wet in the population? Rain affects both sprinkler use and wet grass, so observing sprinkler status carries information about rain. An intervention breaks the incoming rain-to-sprinkler mechanism.

The oracle is P(W=1 | do(S=s)) = sum_r P(W=1 | R=r,S=s)P(R=r). For s=1: 0.3*0.99 + 0.7*0.8 = 0.857. For s=0: 0.3*0.9 + 0.7*0.02 = 0.284. The average causal effect is 0.857-0.284=0.573. Fitted estimates differ due to finite sampling and smoothing. Rain->Sprinkler in this model means rain changes sprinkler use, and that influence is negative. A directed edge does not imply a positive effect.

Variable elimination sums nuisance variables using factors. The intervention implementation removes incoming edges, then fixes the intervention variable using evidence. For a fixed state the replacement prior cancels on normalization. This implementation is for the supplied discrete causal DAG under the stated assumptions.

9. Why an intervention is not an individual counterfactual

Our result averages over the population under a changed mechanism. An individual counterfactual asks what would have happened to a specific case under a different treatment, conditional on what actually happened. It requires abduction, action and prediction in an SCM with a defensible joint model of exogenous variables. The current code does not implement individual counterfactual inference. Say this directly if asked.

10. Five-minute live demonstration

1. Open Overview and name the three separate tasks. Mention Phase 1 scope.

2. Open Reasoning lab. Run a bundled example and distinguish the model decision from its uncalibrated score and dataset label.

3. Open Extraction benchmark. Show measured rule/LR comparison and entity-only ablation. Classify a marked sentence. Explain given-entity evaluation.

4. Open Text -> DAG. Use 'Rain causes sprinkler. Rain causes wet grass. Sprinkler causes wet grass. Wet grass causes rain.' Approve all. Show the rejected cycle. Explain source-order dependence.

5. Keep the three accepted edges. Select synthetic observations, enter your reviewer name and 'Known synthetic generating graph', then explicitly check assumptions. Fit the graph and compare sprinkler interventions.

6. Open Intervention lab. Set sprinkler to 0 and 1. Explain the oracle and the observational difference. End with the limits of the demonstration.

11. Questions examiners may ask

Why add SemEval when Corr2Cause is recommended? We retain Corr2Cause. Its labels evaluate causal reasoning entailment, so it cannot independently validate extraction. SemEval supplies directed cause-effect relation labels and given entity pairs as a separate benchmark. It does not establish performance on scientific/news documents. The faculty's dataset-flexibility rule permits an additional suitable dataset.

Why not use random forest like the reference? Sparse high-dimensional text has a natural linear baseline. The reference's tabular medical preprocessing and tree models solve another problem. We adopt its validation discipline, not an unjustified model checklist.

Does review make the graph causal? No. Review filters predictions and records assumptions. We still need domain knowledge, identification assumptions and suitable data.

Can you estimate CPDs by counting causal mentions? No. Mention frequencies and classifier scores do not describe joint observations of modeled variables.

Why not keep tuning until test F1 improves? That would use test labels for development. New tuning decisions require fresh evaluation or a declared retrospective analysis.

What if an edge has no observations? The fitting contract rejects missing variable columns. A text-derived graph with no measured variables remains a structure proposal.

What does the entity-only ablation prove? It is diagnostic evidence about lexical information, not a proof of complete robustness. Compare results on the same frozen test rows.

What is original about the project? The contribution is the explicit separation and audit of language classification, edge review, graph constraints and Bayesian querying, with a demonstrable bridge using matching observations. Do not claim a new causal-identification theorem or a new state-of-the-art architecture.

12. Preparation for all five members

Each member should explain the complete pipeline, reproduce one result, derive one query, explain one failure case and locate the relevant code. Rehearse a 60-second summary, a five-minute demo and five difficult questions without reading slides. Then fill docs/contributions.md with actual completed work and commit/evidence links. Suggested practice areas may be shared, but all five remain individually accountable for all concepts.

Presentation delivery and speaking script

Timing and suggested speaking roles

This eight-slide talk takes approximately 6 minutes. Allow another 2 minutes only if the panel requests a live demonstration. Rehearse with the script, then speak naturally.

Suggested speaking blocks: Yug slides 1-2, Vidit slides 3-4, Aditya slide 5, Kaavya slide 6, Ishita slides 7-8. These are suggested speaking assignments, not claims of implementation ownership. Every member should understand the complete workflow for the individual viva.

Before presenting, open the deck in slideshow mode, check the display and have the offline synthetic demonstration ready. State the task and metric before giving each score. Pause after the key result and use the script's transition to hand over to the next speaker.

SLIDE 01 / Causal Reasoning

Yug Johri / approximately 30 seconds

Delivery cue: Introduce the team and the project in everyday language.

Good morning. We are Team T094 from Section E. Our project studies how language-based causal claims can be connected to Bayesian networks. We use classical machine learning to evaluate text and predict candidate relations. We then separate those predictions from the assumptions needed for causal queries. Our presentation covers the workflow, datasets, measured results and the observation-versus-intervention example.

Transition: Here is the overall workflow.

SLIDE 02 / System Components

Yug Johri / approximately 40 seconds

Delivery cue: Point to the language evaluation and formal reasoning branches.

The language component evaluates two different tasks. Corr2Cause asks whether a hypothesis follows from a premise. SemEval classifies a directed causal relation between two supplied entities. Candidate edges need human approval and sentence provenance before entering a directed acyclic graph. Matching observations then estimate the Bayesian probability tables. This keeps language predictions, graph validity and causal assumptions separate.

Transition: Vidit will explain the data and model choices.

SLIDE 03 / Datasets and Split Integrity

Vidit Sachan / approximately 40 seconds

Delivery cue: State the split sizes briefly, then focus on leakage prevention.

Corr2Cause uses twelve thousand training rows, with one thousand forty development and one thousand forty-four retained test rows. The extraction extension uses sixty-four hundred training, sixteen hundred development and twenty-seven hundred retained test pairs. We separate exact premise groups for reasoning and sentence groups for extraction. SemEval supplies the entity spans and lacks document identifiers, which limits our evaluation claim. All learned text features fit training data only.

Transition: The next slide shows how those texts become model inputs.

SLIDE 04 / Classical Text Models

Vidit Sachan / approximately 45 seconds

Delivery cue: Explain TF-IDF simply and emphasize development-only selection.

TF-IDF represents texts using weighted words and short phrases. Bigrams preserve cues such as caused by. Balanced Logistic Regression helps account for class imbalance. We compare it with majority prediction, Naive Bayes and simpler feature controls. In extraction, the context model replaces entity strings with entity-one and entity-two markers. We choose regularization using development edge F1, not test labels. The entity-only ablation removes surrounding context.

Transition: Aditya will explain the reasoning results and their main limitation.

SLIDE 05 / Reasoning Results

Aditya Verma / approximately 45 seconds

Delivery cue: Name macro F1 before pointing to the full-input and hypothesis-only bars.

The full-text Corr2Cause model reaches a test macro F1 of zero point seven zero zero. The majority baseline reaches about zero point four six zero. However, the hypothesis-only control reaches zero point six five three without reading the premise. This indicates that wording shortcuts provide substantial signal. We therefore present the model as a classical text-classification baseline and do not treat its score as proof of formal causal understanding.

Transition: Kaavya will now explain the separate extraction result.

SLIDE 06 / Directed Extraction

Kaavya Gala / approximately 45 seconds

Delivery cue: Compare the context model with both controls and explain direction errors.

For directed extraction, context TF-IDF with Logistic Regression reaches edge F1 of zero point eight two zero. Lexical rules reach zero point seven four two, and the entity-only ablation reaches zero point five zero three. Predicting the reverse direction creates both a false positive and a false negative. These results show the value of sentence context on this given-entity task. They do not establish causal truth or automatic entity discovery. Proposed edges still require review before Bayesian fitting.

Transition: Ishita will explain what the Bayesian stage adds.

SLIDE 07 / Observation and Intervention

Ishita Thakur / approximately 50 seconds

Delivery cue: Explain both questions in plain language before referring to formulas.

Observation asks how often the grass is wet among cases where the sprinkler is on. Intervention asks what happens if we force the sprinkler on in the population. Rain affects both sprinkler use and wet grass, so observing sprinkler status also gives information about rain. An intervention breaks the incoming sprinkler mechanism. Our approved graph is a DAG, and matching binary observations estimate its conditional probability tables using BDeu smoothing. Causal interpretation depends on the graph, observed confounders and stable mechanisms.

Transition: The final slide checks those calculations and summarizes our findings.

SLIDE 08 / Results and Conclusions

Ishita Thakur / approximately 50 seconds

Delivery cue: Point to fitted and oracle values, then end with the project contribution.

We validate the Bayesian implementation using six thousand synthetic observations from a known system. With the sprinkler forced on, the exact wet-grass probability is zero point eight five seven and the fitted value is about zero point eight four six. With it off, the values are zero point two eight four and zero point two seven five. The maximum absolute error is approximately zero point zero one one. Our conclusion is that context helps relation classification, while formal causal querying additionally needs reviewed structure, suitable observations and explicit assumptions. The notebook, saved-run verifier and seventeen passing tests make the workflow inspectable. Thank you. We are ready for questions.

Transition: Keep this result slide visible for questions. Offer the live demonstration if requested.

Live-demo script and recovery plan

Say: "I will now show the reviewed-graph path using a synthetic system with known assumptions. This validates the calculation, not a real-world causal discovery claim."

In Text to DAG, enter: Rain causes sprinkler. Rain causes wet grass. Sprinkler causes wet grass. Wet grass causes rain. Approve the candidates, show the cycle rejection and keep the three accepted edges. Say: "The final edge closes a cycle. Our static Bayesian network requires a DAG, so the graph builder rejects it. Source order affects this decision."

Select synthetic observations, enter your actual reviewer name and the rationale "Known synthetic generating graph", and explicitly check the stated assumptions. Fit the network. Set sprinkler to one and query wet grass. Say: "The fitted intervention probability is close to the known value 0.857. Its small difference reflects sampled observations and smoothing."

If the internet fails, use the bundled offline inference and synthetic example. If the live app fails, return to slide 8 and explain the saved oracle comparison. Do not invent a result or claim the live demo succeeded. If time runs short, skip the demo and move to the key findings.

Answering questions professionally

Listen to the complete question. Begin with a direct answer, then connect it to one equation, code module or measured result. If asked whether the model establishes causation, say: "The text model identifies candidate relations. Causal effects follow only under the supplied graph and identification assumptions." If you do not know an answer, state what you do know and identify the additional experiment or assumption needed. Avoid guessing.

Before the viva, every member should rehearse the observation-versus-intervention explanation, derive 0.857 from the oracle, explain why a reversed edge counts as both an FP and FN, and explain why the hypothesis-only control limits our reasoning claim.